

Introduction to Physiology & the Chemical Level

What physiology studies — and the one idea the whole course is built on.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Say what physiology studies, and how it is different from anatomy
- Explain homeostasis, and why the body defends a set point
- Name the three parts of a feedback loop and say what each one does
- Say why water, pH, and the four macromolecules matter to a nurse
- Put the levels of organization in order, from atom to organism

1 Anatomy asks what. Physiology asks why.

Anatomy is the parts list. Physiology is the instruction manual.

Anatomy tells you the heart has four chambers. It tells you there is a wall between the two sides.

Physiology tells you *why* that wall has to be there. It also tells you what happens to a patient when the wall has a hole in it.

All semester, when you meet a new structure, ask one question: **what is this shaped to do?**

THINK OF IT THIS WAY

Anatomy is knowing a building has sprinklers. Physiology is knowing what turns them on, and what gets soaked when they fail.

2 Homeostasis — the idea everything hangs on

Your body holds dozens of values steady, every second, and you never notice.

Body temperature stays near 37°C. Blood pH stays near 7.4. Blood sugar, calcium, sodium, and oxygen all stay inside narrow ranges.

That steady inside state is **homeostasis**. Keeping it steady is most of what your organ systems do all day.

The value the body aims for is called the **set point**. When you drift away from it, something switches on to pull you back.

Almost every disease you will study is a loop like this that stopped working. That is why this idea comes first.

3 How a feedback loop works

Every loop has the same three parts. Learn them once and you will spot them all semester.

- 1 **Receptor.** It notices the change. Something senses that you are too warm.
- 2 **Control center.** Usually the brain. It compares what is happening to the set point, then decides.
- 3 **Effector.** It does the fixing. Sweat glands, blood vessels, a muscle, or a gland.

The response works *against* the change. So we call it **negative feedback**.

Too warm turns on cooling. Too cool turns on warming. Here, negative means opposing. It does not mean bad.

BUILD A HOMEOSTATIC LOOP — THEN BREAK IT

SET POINT 37.0°C

Negative feedback: ON

Drag the temperature away from the set point. Watch the loop pull it back.

Now press **Break the loop** and drag again. Nothing corrects it.

That is a fever that will not come down. It is a blood sugar that will not fall. It is a blood pressure that will not settle.

Positive feedback is the opposite, and it is rarer. The response *builds on* the change instead of fighting it. The body uses it when it wants something finished fast. Blood clotting and labor contractions are the two classic examples. It always takes an outside event to stop it.

4 The chemistry you actually need

You are not becoming a chemist. You need four things that explain a lot of physiology.

- **Water.** About 60% of you. It dissolves things, carries them, and warms and cools slowly. That slowness is why you can hold 37°C.
- **pH.** Blood sits between 7.35 and 7.45. That window is tight. Outside it, proteins change shape and stop working. This is why acid-base problems are an emergency in nursing.
- **Ions.** Sodium, potassium, calcium. These are charged particles. Their movement makes every nerve signal and every heartbeat.
- **The four macromolecules.** Carbohydrates are quick fuel. Lipids are stored fuel and membranes. Proteins are structure and almost all the machinery. Nucleic acids are the instructions.

Proteins deserve extra attention. Enzymes are proteins. Receptors are proteins. Ion channels are proteins.

A protein only works in the right shape. That is why a high fever or a pH swing is dangerous. Change the shape and you break the job it does.

5 Levels of organization

The course is a climb. Each level only makes sense once you know the one below it.

Atoms → molecules → cells → tissues → organs → organ systems → organism.

That is also the shape of the semester. We start with chemistry and cells because everything after them is built from them.

When the heart comes up in Module 10, you will need the ion movement from Module 4 to understand why it beats at all. It really does stack.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A patient gets too warm and starts to sweat. In the feedback loop, what is the sweat gland?

- a) The receptor
- b) The control center
- c) The effector
- d) The set point

2. Why is negative feedback called 'negative'?

- a) Because the result is harmful
- b) Because the response works against the change
- c) Because it only works when a value drops
- d) Because it shuts the organ system down

3. Blood clotting speeds itself up until the bleeding stops. What kind of feedback is that?

- a) Negative feedback
- b) Positive feedback
- c) A set point
- d) Denaturation

4. A patient's blood pH drops to 7.1. Why is that dangerous?

- a) Water can no longer dissolve anything
- b) Proteins change shape and stop working
- c) The body runs out of carbohydrates
- d) Cells can no longer divide

5. Which list puts the levels of organization in the right order?

- a) Cell, atom, tissue, organ
- b) Atom, molecule, cell, tissue, organ
- c) Molecule, atom, organ, tissue
- d) Tissue, cell, molecule, atom

Answer key

1. **The effector**

Right. The receptor notices, the control center decides, and the effector acts. Sweating is the action.

2. **Because the response works against the change**

Yes. Too warm turns on cooling, and too cool turns on warming. Negative means opposing.

3. **Positive feedback**

Correct. The response builds on the change, and an outside event, the sealed vessel, ends it.

4. **Proteins change shape and stop working**

Yes. Enzymes, receptors, and channels are all proteins. Shape is what makes them work.

5. **Atom, molecule, cell, tissue, organ**

Right. Smallest to largest, and each level is built from the one before it.

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